

## Final Exam – Medicinal Plants (Lectures 5–12)

Essay & Completion Based Exam

Time: 3 Hours

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### Section A – Essay & Completion Questions

1. Write short notes on Senna leaves including active constituents, mechanism of action, uses, and adverse effects.
2. Explain why Senna should not be used for long-term treatment of constipation.
3. Describe the toxic components of Boldo leaves and their clinical significance.
4. Discuss the role of Artichoke leaves in management of hyperlipidemia.
5. Compare Senna and Uva Ursi in terms of site of action and therapeutic use.
6. Complete: Arbutin is converted into \_\_\_\_\_ in \_\_\_\_\_ urine to exert antibacterial effect.
7. Explain why Buchu is contraindicated in kidney inflammation.
8. Write short notes on Dandelion as a liver tonic.
9. Complete: Guava leaves are rich in \_\_\_\_\_ which makes them useful in \_\_\_\_\_.
10. Differentiate between dry cough and productive cough with suitable medicinal plants.
11. Describe the pharmacological action of Eucalyptus oil in respiratory conditions.
12. Explain the toxicity associated with oral ingestion of Eucalyptus oil.
13. Write short notes on Ivy leaves as expectorant agents.
14. Complete: Ivy leaves contain \_\_\_\_\_ which reduce surface tension of bronchial mucus.
15. Compare Guava and Ivy regarding mechanism of action in cough.
16. Discuss the stimulant effects of Tea leaves.
17. Explain the role of tannins present in Tea.
18. Write short notes on Hyoscyamus toxicity.
19. Complete: Anticholinergic toxicity is characterized by \_\_\_\_\_, \_\_\_\_\_ and \_\_\_\_\_.
20. Discuss the therapeutic uses of Jaborandi leaves.

21. Explain the pharmacological action of Coca leaves.
22. Write short notes on Digitalis leaves including mechanism of action.
23. Explain why Digitalis has a narrow therapeutic index.
24. Complete: Digoxin is mainly obtained from \_\_\_\_ species.
25. Discuss the clinical importance of monitoring electrolytes during Digitalis therapy.
26. Write short notes on Ginkgo leaves and their therapeutic uses.
27. Explain the interaction between Ginkgo and anticoagulant drugs.
28. Define plant taxonomy and state its importance in pharmacognosy.
29. Complete: The binomial system of nomenclature was introduced by \_\_\_\_.
30. Differentiate between monocots and dicots based on morphological characters.
31. Explain the difference between gymnosperms and angiosperms.
32. Complete: Plants with naked seeds are classified as \_\_\_\_.
33. Write short notes on Myrtaceae family with example.
34. Write short notes on Asteraceae family with example.
35. Explain why Uva Ursi is not suitable for chronic use.
36. Complete: The toxic component of Boldo is \_\_\_\_.
37. Discuss adverse effects of chronic laxative abuse.
38. Explain the role of saponins in respiratory therapy.
39. Complete: The major volatile oil in Eucalyptus is \_\_\_\_.
40. Write short notes on Plantaginaceae family with example.
41. Explain why hypokalemia increases Digitalis toxicity.
42. Complete: The antibacterial metabolite released from arbutin is \_\_\_\_.
43. Discuss the importance of proper plant identification in medicinal use.
44. Write short notes on Leguminosae family with example.
45. Complete: The family of Guava is \_\_\_\_.
46. Explain why Ivy is contraindicated in infants.

47. Discuss clinical signs of anticholinergic poisoning.
48. Complete: The main cardiac glycoside in *Digitalis lanata* is \_\_\_\_\_.
49. Write short notes on therapeutic importance of bile secretion.
50. Explain the difference between hepatoprotective and choleretic actions.

## Section B – Model Answers

1. Senna contains anthraquinone glycosides which stimulate colonic peristalsis and inhibit water reabsorption, producing laxative effect.
2. Long-term use of senna causes electrolyte imbalance, dependency and colonic atony.
3. Boldo contains ascaridole, a toxic monoterpene responsible for hepatotoxicity.
4. Artichoke increases bile secretion through cynarin, enhancing cholesterol excretion.
5. Senna acts on the colon as laxative, while Uva Ursi acts in urinary tract as antiseptic.
6. Hydroquinone – alkaline
7. Buchu irritates renal tissue leading to worsening kidney inflammation.
8. Dandelion acts as mild diuretic and hepatoprotective agent.
9. Tannins – diarrhea and dry cough
10. Dry cough is treated with guava, productive cough with ivy.
11. Eucalyptus oil acts as decongestant and expectorant.
12. Oral toxicity leads to nausea, vomiting and respiratory depression.
13. Ivy contains saponins that reduce mucus viscosity and aid expectoration.
14. Triterpenoid saponins
15. Guava is astringent while ivy is expectorant.
16. Tea contains xanthines causing CNS stimulation.
17. Tannins cause astringency and reduce secretion.
18. Hyoscyamus causes anticholinergic toxicity.
19. Dry mouth, mydriasis, tachycardia
20. Jaborandi contains pilocarpine used in glaucoma.
21. Coca leaves act as CNS stimulant and reduce fatigue.
22. Digitalis increases force of cardiac contraction by increasing intracellular calcium.
23. Narrow margin between therapeutic and toxic doses.
24. Digitalis lanata
25. Hypokalemia increases digitalis toxicity.

26. Ginkgo improves cerebral circulation and memory.
27. Ginkgo increases bleeding risk with anticoagulants.
28. Plant taxonomy is classification of plants for proper identification.
29. Linnaeus
30. Monocots have one cotyledon; dicots have two.
31. Gymnosperms have naked seeds; angiosperms have covered seeds.
32. Gymnosperms
33. Myrtaceae includes Guava.
34. Asteraceae includes Artichoke.
35. Long-term use causes hydroquinone toxicity.
36. Ascaridole
37. Causes dependency and electrolyte imbalance.
38. Saponins reduce surface tension of mucus.
39. 1,8-Cineole
40. Plantaginaceae includes Digitalis.
41. Hypokalemia increases glycoside binding.
42. Hydroquinone
43. Prevents adulteration and toxicity.
44. Leguminosae includes Senna.
45. Myrtaceae
46. Causes GI irritation in infants.
47. Hallucinations, dry mouth, tachycardia.
48. Digoxin
49. Bile aids digestion and cholesterol excretion.
50. Hepatoprotective protects liver cells, choleretic increases bile flow.